

Figure 4: Gateway Group – DualWAN



Figure 5: Floating rule for dual WAN

configured with two Internet connections, one each on the WAN and OPT1, go ahead and create a dual WAN group by going to **System – Routing – Gateway Groups**. Then configure a dualWAN group for WAN with Tier1 (highest priority) and OPT1 with Tier2 (second highest priority).

3. Add/configure the firewall rules required for a dual WAN set-up. Use the DualWAN group created in Step 2 to create new floating rules.

- a. Go to the floating rule creation screen menu: **Firewall – Rules – Floating**. Add new floating rule as per the screenshot shown in Figure 5. Modify the existing firewall rules by using DualWAN in place of WAN.
- b. Modify existing LAN to any rule - which is created by pfSense automatically at the time of installation. This rule allowed all traffic originating from LAN towards all destinations on all services (ports) using WAN as the gateway.

Then, open **Open Firewall – Rules – LAN – Gateway box**, click the **Display Advanced** button and change existing WAN gateway to DualWAN gateway created earlier in Step 2.

For creating additional firewall rules, use the DualWAN gateway instead of WAN.

4. **Reconfigure the Squid Proxy service:** Two changes are required in Squid Proxy configuration.

- a. Select 'LAN' and 'loopback' proxy interfaces under **Squid General Settings**.

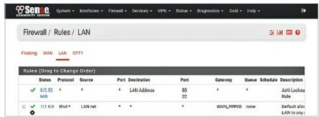


Figure 6: Default 'LAN net' rule

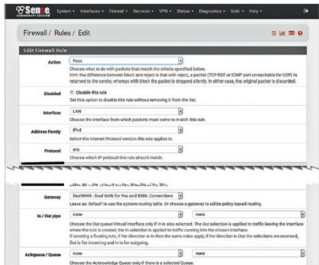


Figure 7: Change gateway to DualWAN



Figure 8: DNS server configuration

- b. Click **Show Advanced Options** and add 'tcp_outgoing_address 127.0.0.1;' string at start of the 'Integration' box.
5. **Configure the default gateway auto failover:** The pfSense website mentions that, by default, traffic using a proxy such as Squid will bypass policy routing and use the default route for traffic at all times. It also bypasses the expected outbound NAT and leaves via the WAN IP address, directly.
Further, policy routing traffic from the firewall itself is not currently possible. Failover can be achieved in many cases by using the default gateway switching under **System > Advanced** on the **Miscellaneous** tab.
Proceed to **System – Advanced – Miscellaneous** tab and select the 'Default Gateway Switching' box.
 6. **Best practices when configuring DNS servers:** Typically, WAN and OPT1 connections will be from different ISPs. Thus, these connections will have different DNS settings belonging to corresponding ISPs.